

XUEPENG FU

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EDUCATION

Shanghai Jiao Tong University Sept. 2018 - Dec. 2023
Ph.D. in Ocean Engineering (Advisor: Shixiao Fu, Outstanding thesis award) *Shanghai, China*
Ocean University of China Sept. 2014 - July 2018
B.Eng. in Ocean Engineering (Advisor: Weiping Huang, Outstanding thesis award) *Qingdao, China*

ACADEMIC EMPLOYMENT

Postdoctoral Researcher in Mechanical Engineering Apr. 2024 - Present
National Laboratory of the Rockies, Golden, CO, USA
Research Assistant in Ocean Engineering Jan. 2024 - Mar. 2024
Shanghai Jiao Tong University, Shanghai, China

RESEARCH INTERESTS

Fluid-structure interaction (especially vortex-induced vibration) in offshore oil & gas and renewable energy systems, investigated through experimental, numerical, and data-driven approaches.

JOURNAL ARTICLES * : CORRESPONDING AUTHOR

24. **X. Fu***, F. Driscoll, R. Fao, C. Kenny, K.P. Griffin, M. Murphy & S. Lambert. "Performance evaluation of an offshore wave measurement buoy in monochromatic waves." (Under Revision, *Ocean Engineering*)
23. **X. Fu**, S. Fu*, C. Liu, J. Shen, & Z. Niu. (2026) "Hydrodynamic properties of a flexible pipe in bidirectionally sheared flows undergoing vortex-induced vibration." *Journal of Fluid Mechanics*, 1034, A44.
22. **X. Fu***, G. Vijayakumar, B. Lee, & M. Sprague (2026). "Numerical simulation of vortex-induced vibration response of a single IEA 10-MW wind turbine blade." *Journal of Fluids and Structures*, 143, 104537.
21. Z. Niu, S. Fu*, **X. Fu**, X. Liu, M. Zhang, J. Shen & P. Deng (2026). "Experimental investigation on vortex-induced vibration of a free-hanging flexible riser under different bottom clump weights." *Ocean Engineering*, 353, 124778.
20. D. Wang, Q. Hu, X. Xing, W. Hong, J. Liu, **X. Fu** & C. Yang* (2026). "Thinking like clinicians: Ranking-based multi-instance learning for PPG-based hemodynamic fluctuation detection." *Pattern Recognition*, 174, 112981.
19. J. Shen, S. Fu*, **X. Fu**, P. Deng, Z. Niu & T. Moan (2026). "A hybrid experimental investigation of structural parameter effects on 2-DOF vortexinduced vibrations of a circular cylinder." *Journal of Ocean Engineering and Science*, 11(1), 191-203.
18. J. Shen, S. Fu*, **X. Fu**, T. Moan & S. Sævik (2025). "Flow-induced vibration of twin-pipe model with varying mass and damping: A study using virtual physical framework." *Journal of Fluids and Structures*, 139, 104438.
17. J. Shen, P. Deng, S. Fu*, **X. Fu**, L. Song & Y. Xu (2025). "Enhanced vortex-induced vibration prediction of marine risers using excitation coefficient databases at high Reynolds numbers." *Journal of Fluids and Structures*, 138, 104406.

16. **X. Fu**, S. Fu*, Z. Niu, B. Zhao, J. Shen & P. Deng (2025). "A validated fluid-structure interaction simulation model for vortex-induced vibration of a flexible pipe in steady flow." *Marine Structures*, 104, 103895.
15. **X. Fu**, S. Fu*, Z. Han, Z. Niu, M. Zhang & B. Zhao (2025). "Numerical simulations of 2-DOF vortex-induced vibration of a circular cylinder in two and three dimensions: A comparison study." *Journal of Ocean Engineering and Science*, 10(4), 395-410.
14. **X. Fu**, S. Fu*, C. Liu, M. Zhang & Q. Hu (2024). "Data-driven approach for modeling Reynolds stress tensor with invariance preservation." *Computers & Fluids*, 274, 106215.
13. B. Zhao, S. Fu*, P. Deng, M. Zhang, Y. Bai & **X. Fu** (2024). "Frequency-domain prediction method for vortex/wake-induced vibrations of double flexible risers in tandem arrangements." *Ocean Engineering*, 297, 116942.
12. **X. Fu**, S. Fu*, M. Zhang, H. Ren, B. Zhao & Y. Xu (2024). "Vortex-induced vibration of flexible pipe under oscillatory sheared flow." *Physical Review Fluids*, 9(1), 014604.
11. **X. Fu**, S. Fu*, M. Zhang, Y. Xu, H. Ren & T. Sun (2024). Mean drag force of flexible riser under bidirectionally sheared flow. *Journal of Shanghai Jiao Tong University*, 58(11), 1006-2467. (In Chinese)
10. Z. Yang, Y. Xu*, J. Jing, **X. Fu**, B. Wang, H. Ren, M. Zhang & T. Sun (2023). "Investigation of physics-informed neural networks to reconstruct a flow field with high resolution." *Journal of Marine Science and Engineering*, 11(11), 2045.
9. B. Zhao, M. Zhang*, S. Fu, **X. Fu**, H. Ren & Y. Xu (2023). "Drag coefficients of double unequal-diameter flexible cylinders in tandem undergoing vortex/wake-induced vibrations." *Ocean Engineering*, 270, 113642.
8. B. Zhao, M. Zhang*, S. Fu, **X. Fu**, T. Sun & B. Song (2023). "Experimental investigation on vortex/wake-induced force of double unequal-diameter cylinders in tandem." *Physics of Fluids*, 35(5), 055134.
7. H. Ren, S. Fu*, B. Zhao, M. Zhang, Y. Xu, J. Shen, **X. Fu** & J. Huang (2022). "Hydrodynamic force model for flexible pipe based on energy competition and applications into flow induced vibration prediction in uniform flow." *Marine Structures*, 86, 103291.
6. **X. Fu**, S. Fu*, H. Ren, W. Xie, Y. Xu, M. Zhang, Z. Liu & S. Meng (2022). "Experimental investigation of vortex-induced vibration of a flexible pipe in bidirectionally sheared flow." *Journal of Fluids and Structures*, 114, 103722.
5. **X. Fu**, S. Fu*, M. Zhang, Z. Han, H. Ren, Y. Xu & B. Zhao (2022). "Frequency capture phenomenon in tandem cylinders with different diameters undergoing flow-induced vibration." *Physics of Fluids*, 34(8), 085120.
4. **X. Fu**, M. Zhang*, S. Fu, B. Zhao, H. Ren & Y. Xu (2022). "On the study of vortex-induced vibration of a straked pipe in bidirectionally sheared flow." *Ocean Engineering*, 266, 112945.
3. H. Song, W. Huang*, **X. Fu**, H. Yan & S. Chang (2021). "Empirical model of the wake-induced lift force on a cylinder with low mass ratio." *Marine Structures*, 80, 103081.
2. H. Ren, M. Zhang*, Y. Wang, Y. Xu, S. Fu, **X. Fu** & B. Zhao (2020). "Drag and added mass coefficients of a flexible pipe undergoing vortex-induced vibration in an oscillatory flow." *Ocean Engineering*, 210, 107541.
1. **X. Fu** & W. Huang* (2018). Experimental study of out-of-plane motion of steel catenary riser in rigid mode. *The Ocean Engineering*, 36(5): 114-120. (In Chinese)

3. Y. Bai, M. Zhang, S. Fu, Y. Chu, **X. Fu**, H. Ren, & Y. Xu (2024). An experimental investigation on vortex-induced vibration of double risers in tandem under bidirectionally sheared flow. In *International Conference on Offshore Mechanics and Arctic Engineering*, 87844, V006T08A019.
2. **X. Fu**, Y. Xu, M. Zhang, H. Ren, B. Zhao, & S. Fu (2020). Numerical simulation of vortex-induced vibration of two tandem cylinders with different diameters under uniform Flow. In *International Conference on Offshore Mechanics and Arctic Engineering*, 84409, V008T08A034a.
1. B. Zhao, M. Zhang, Y. Xu, H. Ren, **X. Fu**, S. Fu, & C. Li (2020). Experimental study on interference response characteristics of triple flexible risers under uniform flow. In *International Ocean and Polar Engineering Conference*. ISOPE-I-20-2174.

PATENTS

4. **X. Fu**, W. Huang, C. Guo, Q. Wu, T. Yang, A. Liu, H. Song (2018). A method for determining the vibration response frequency of tandem cylinders in the wake. CN108168684A. (Chinese Patent)
3. **X. Fu**, M. Zhang, Q. Wu, Z. Jiang, C. Guo, A. Chang, W. Ma (2017). A movable multifunctional modular floating island group. CN106494587B. (Chinese Patent)
2. H. Ren, S. Fu, **X. Fu**, J. Shen, M. Zhang, Y. Xu, T. Hu, B. Song, T. Sun (2025). Experimental setup for vortex-induced vibration of a riser in a bidirectional shear flow with flexible transmission. CN116625617A. (Chinese Patent)
1. Ren, Y. Xu, S. Fu, M. Zhang, **X. Fu**, J. Shen, B. Song, T. Sun (2025). Experimental setup for vortex-induced vibration of a riser in a bidirectional shear flow with flexible transmission. CN116519240A. (Chinese Patent)

ACADEMIC TALKS

6. “Experimental investigation of frequency-domain wave energy estimation uncertainty via large-amplitude motion platform”, *Offshore Technology Conference 2026*, May, 2026, Houston, USA.
5. “Uncertainty estimation of wave buoy measurement via Large Amplitude Motion Platform”, *NREL Center for Energy Conversion and Storage Systems Tech Talk*, Oct., 2025, Virtual.
4. “Vortex-induced vibration of a flexible cylinder under bidirectionally sheared flow”, *76th Annual Meeting of the Division of Fluid Dynamics*, Nov., 2023, Washington DC, USA.
3. “Investigation on layout optimization of helical strakes on marine riser”, *42nd International Conference on Ocean, Offshore and Arctic Engineering*, June, 2023, Melbourne, Australia.
2. “Experimental investigation of vortex-induced vibration of a flexible pipe with helical strakes in oscillatory flow”, *Academic Conference of Chinese Society of Naval Architects in 2021*, Oct., 2021, Kunming, China. (Outstanding Presentation, Top 2%)
1. “Numerical simulation of vortex-induced vibration of two tandem cylinders with different diameters under uniform flow”, *39th International Conference on Ocean, Offshore and Arctic Engineering*, Aug., 2020, Virtual.

PROFESSIONAL SERVICE

2. Early Career Editorial Board Member, *Journal of Fluids and Structures* Dec. 2025 - Present
1. Reviewer for *J. Fluids Struct.*, *Mar. Struct.*, *Ocean Eng.*, *Appl. Ocean Res.*, *Int. Commun. Heat Mass Transfer*, *Phys. Fluids*, *OMAE*, *ISOPE*, et al.

HONORS & AWARDS

Key Contributor Award at National Laboratory of the Rockies	2025
National Outstanding Doctoral Dissertation in Ocean Engineering	2025
Outstanding Doctoral Dissertation at Shanghai Jiao Tong University	2023
Weichai Power Scholarship	2022
Outstanding Undergraduate Graduates at Ocean University of China	2018
Outstanding Bachelor Thesis at Ocean University of China	2018
Science and Technology Innovation Scholarship at Ocean University of China	2015, 2016, 2017, 2018

PROJECTS EXPERIENCE

1. Primary Research of Autonomous Wave Tank Experimental System, Open R&D Program, Department of Energy, USA, 09/2025 - 02/2026 (PI)
2. Uncertainty Quantification of Wave Resource Characterization, Department of Energy, USA, 04/2024 - Now (Technical Lead)
3. PacWave: Offshore Wave Energy Test Site, Department of Energy, USA, 04/2024 - Now (Subtask Technical Lead)
4. High-Fidelity Modeling, Department of Energy, USA, 04/2024 – 09/2025 (Subtask Technical Lead)
5. FIFERS - Flow Induced Force and its Effects on Risers, Joint Industry Program, DNV, Aker Solutions, Equinor, CNOOC, CPR Subsea, ABS, 10/2020 - 12/2023 (Technical Lead)
6. The Couplings Among Multi-Oceanic Fields, Multi-body and Multi-scale and Their Effects on the Performance and Safety of Ocean Engineering Equipment, Fundamental Science Center Project, NSFC, China, 01/2021 - 12/2023 (Subtask Technical Lead)
7. Theory of Fluid-Structure Interaction Response Analysis for Ultra-Slender Flexible Structures, Major Program, NSFC, China, 09/2018 - 12/2020 (Subtask Technical Lead)

REFERENCES

1. Name: Frederick Driscoll (NLR)
Title: Principal Engineer in Water Power at National Laboratory of the Rockies
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2. Name: Ganesh Vijayakumar (NLR)
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3. Name: Chang Liu (UConn)
Title: Assistant Professor at University of Connecticut
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4. Name: Shixiao Fu (SJTU, NTNU)
Title: Professor and Department Dean at Shanghai Jiao Tong University
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